But, phosphate concentration is very low in blood, thus, phosphate buffer, plays a major role as an intracellular buffer in red blood cell and other types of cells where their concentrations are higher than in blood and interstitial fluid. The sodium salts of phosphoric acid also act as buffer system.

Protein buffer system

Plasma proteins are responsible for protein buffer system. The buffering capacity of proteins depends upon the pK_a of ionisable group of aminoacid side chains. Histidine residue plays a vital role as buffering agent because its imidazole group pK_a value is 6.7 and it is the more effective contributor for protein buffer system. The plasma proteins are responsible for the 2% buffering capacity of plasma. At blood pH 7.4, proteins exist as anions (Pr^-) serving as conjugate base. After accepting H^+ ions it is converted into weak acid (HPr). Thus, buffering action of proteins is due to the following dissociation reaction:

$$HPr \rightleftharpoons Pr^- + H^+$$
$$\text{Protein-H} \rightleftharpoons \text{Protein}^- + H^+$$

Hemoglobin buffer systems

Hemoglobin present in erythrocytes also plays an important role as buffering agent. It mainly buffers the acids produced during gaseous transport between lungs and tissues.

$$HHb \rightleftharpoons Hb^- + H^+$$

At tissue levels, H^+ ions released from carbonic acid bind with haemoglobin and help in the transport of CO_2 as HCO_3^-. In lungs, as haemoglobin combines with oxygen, it releases H^+ ions, which in turn bind with HCO_3^- to form carbonic acid. Carbonic acid then dissociates into CO_2 and water. Then, CO_2 is exhaled. Thus haemoglobin acts as a buffer system.

1.5.6 Acid-Base Balance

The normal pH of biological fluid is maintained in a narrow range. For example, the pH of the blood is maintained between 7.35-7.40, i.e. slightly alkaline. The changes in pH range will affect metabolic functions e.g. denaturation of proteins, enzyme activity etc. Thus, maintenance of pH is vital for normal physiological and biochemical functions of the body. The change of pH is due to the change of acid-base concentrations in the cell and biological fluids. Hence, the control and maintenance of acid-base balance is essential for the maintenance of pH.

Regulation of Acid-Base balance

The acid-base balance in the body is maintained by buffer system along with the functions of lungs and kidney.

Role of Lung

The first line of defense maintenance of pH is the control of extracellular concentrations of CO_2 and bicarbonate ions by the lungs. An increase in ventilation removes CO_2 from extracellular fluid which, in turn reduces

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